

Name of Student: Vyankatesh Rohokale	PRN: 26070122518
Semester: III	Year : AY 2026-27
Subject: Programming with Java	
ASSIGNMENT No : 20	
TITLE : JDBC INSERT, UPDATE, and DELETE Operations	

Problem Statement

Develop a Java application to establish connectivity with a database and perform INSERT, UPDATE, and DELETE operations.

Learning Objectives

To perform database modification operations using JDBC.

Programme

1. Create a Java application to manage **employee records** using CRUD operations

```
import java.sql.Connection;
import java.sql.DriverManager;
import java.sql.PreparedStatement;
import java.sql.ResultSet;
import java.util.Scanner;

public class EmployeeCRUD {

    static Connection con;

    static void insertEmployee(Scanner sc) throws Exception {
        System.out.print("Enter Employee ID: ");
        int id = sc.nextInt();
        sc.nextLine();

        System.out.print("Enter Employee Name: ");
        String name = sc.nextLine();

        System.out.print("Enter Employee Salary: ");
        double salary = sc.nextDouble();
```

```
String query = "INSERT INTO employee VALUES (?, ?)";
```

```
PreparedStatement ps = con.prepareStatement(  
    "INSERT INTO employee (e_id, e_name, e_sal) VALUES  
(?, ?, ?)"  
);
```

```
ps.setInt(1, id);  
ps.setString(2, name);  
ps.setDouble(3, salary);
```

```
ps.executeUpdate();
```

```
System.out.println("Employee inserted successfully");
```

```
ps.close();  
}
```

```
static void displayEmployees() throws Exception {  
    String query = "SELECT * FROM employee";
```

```
PreparedStatement ps = con.prepareStatement(query);  
ResultSet rs = ps.executeQuery();
```

```
System.out.println("\nEmployee Records");  
System.out.println("-----");
```

```
while (rs.next()) {  
    System.out.println("Employee ID: " +  
rs.getInt("e_id"));  
    System.out.println("Employee Name: " +  
rs.getString("e_name"));  
    System.out.println("Employee Salary: " +  
rs.getDouble("e_sal"));  
    System.out.println("-----");  
}
```

```
rs.close();  
ps.close();  
}
```

```
static void updateEmployee(Scanner sc) throws Exception {  
    System.out.print("Enter Employee ID to update: ");  
    int id = sc.nextInt();  
    sc.nextLine();
```

```
System.out.print("Enter New Employee Name: ");  
String name = sc.nextLine();
```

```
System.out.print("Enter New Employee Salary: ");  
double salary = sc.nextDouble();
```

```
String query = "UPDATE employee SET e_name = ?, e_sal  
= ? WHERE e_id = ?";
```

```
PreparedStatement ps = con.prepareStatement(query);
```

```
ps.setString(1, name);  
ps.setDouble(2, salary);  
ps.setInt(3, id);
```

```
int rows = ps.executeUpdate();
```

```
if (rows > 0) {  
    System.out.println("Employee updated  
successfully");  
} else {  
    System.out.println("Employee not found");  
}
```

```
ps.close();  
}
```

```
static void deleteEmployee(Scanner sc) throws Exception {  
    System.out.print("Enter Employee ID to delete: ");  
    int id = sc.nextInt();
```

```
String query = "DELETE FROM employee WHERE e_id = ?";
```

```
PreparedStatement ps = con.prepareStatement(query);
```

```
ps.setInt(1, id);
```

```
int rows = ps.executeUpdate();
```

```
if (rows > 0) {  
    System.out.println("Employee deleted  
successfully");  
} else {  
    System.out.println("Employee not found");  
}
```

```
ps.close();  
}
```

```
public static void main(String[] args) throws Exception {
```

```
Class.forName("com.mysql.cj.jdbc.Driver");
```

```
String db = "jdbc:mysql://localhost:3306/employee";
String user = "root";
String password = "admin@123";
```

```
con = DriverManager.getConnection(db, user, password);
```

```
Scanner sc = new Scanner(System.in);
```

```
int choice;
```

```
do {
    System.out.println("\n==== Employee CRUD =====");
    System.out.println("1. Insert Employee");
    System.out.println("2. Display Employees");
    System.out.println("3. Update Employee");
    System.out.println("4. Delete Employee");
    System.out.println("5. Exit");
```

```
    System.out.print("Enter your choice: ");
    choice = sc.nextInt();
```

```
    try {
        switch (choice) {
            case 1:
                insertEmployee(sc);
                break;
```

```
            case 2:
                displayEmployees();
                break;
```

```
            case 3:
                updateEmployee(sc);
                break;
```

```
            case 4:
                deleteEmployee(sc);
                break;
```

```
            case 5:
                System.out.println("Exiting...");
                break;
```

```
            default:
                System.out.println("Invalid choice");
```

```
        }
    } catch (Exception e) {
```

```

        System.out.println("Error: " +
e.getMessage());
    }

    } while (choice != 5);

    con.close();
    sc.close();
}
}

```

```

piyushrohokale@Piyushs-MacBook-Air JDBC % java -cp "src:lib/mysql-connector-j-26.7.0.jar" EmployeeCRUD
5. Update Employee
4. Delete Employee
5. Exit
Enter your choice: 1
Enter Employee ID: 12
Enter Employee Name: egverbh
Enter Employee Salary: 123
Employee inserted successfully

===== Employee CRUD =====
1. Insert Employee
2. Display Employees
3. Update Employee
4. Delete Employee
5. Exit
Enter your choice: 2

Employee Records
-----
Employee ID: 1
Employee Name: vyankatesh
Employee Salary: 12345.0
-----
Employee ID: 12
Employee Name: egverbh
Employee Salary: 123.0
-----

```

2. Create a Java application to manage **student records** using **CRUD operations**. Perform Create, Read, Update, and Delete operations on student details such as roll number, name, course, and marks using a database.

```

import java.sql.Connection;
import java.sql.DriverManager;
import java.sql.PreparedStatement;
import java.sql.ResultSet;
import java.util.Scanner;

public class StudentCRUD {

    static Connection con;

```

```
static void insertStudent(Scanner sc) throws Exception {  
    System.out.print("Enter Roll Number: ");  
    int rollNo = sc.nextInt();  
    sc.nextLine();
```

```
    System.out.print("Enter Student Name: ");  
    String name = sc.nextLine();
```

```
    System.out.print("Enter Course: ");  
    String course = sc.nextLine();
```

```
    System.out.print("Enter Marks: ");  
    double marks = sc.nextDouble();
```

```
    String query = "INSERT INTO student VALUES  
(?, ?, ?, ?)";
```

```
    PreparedStatement ps = con.prepareStatement(query);
```

```
    ps.setInt(1, rollNo);  
    ps.setString(2, name);  
    ps.setString(3, course);  
    ps.setDouble(4, marks);
```

```
    ps.executeUpdate();
```

```
    System.out.println("Student inserted successfully");
```

```
    ps.close();  
}
```

```
static void displayStudents() throws Exception {  
    String query = "SELECT * FROM student";
```

```
    PreparedStatement ps = con.prepareStatement(query);  
    ResultSet rs = ps.executeQuery();
```

```
    System.out.println("\nStudent Records");  
    System.out.println("-----");
```

```
    while (rs.next()) {  
        System.out.println("Roll Number: " +  
rs.getInt("roll_no"));  
        System.out.println("Name: " +  
rs.getString("name"));  
        System.out.println("Course: " +  
rs.getString("course"));
```

```
        System.out.println("Marks: " +  
rs.getDouble("marks"));  
        System.out.println("-----");  
    }
```

```
    rs.close();  
    ps.close();  
}
```

```
static void updateStudent(Scanner sc) throws Exception {  
    System.out.print("Enter Roll Number to update: ");  
    int rollNo = sc.nextInt();  
    sc.nextLine();
```

```
    System.out.print("Enter New Name: ");  
    String name = sc.nextLine();
```

```
    System.out.print("Enter New Course: ");  
    String course = sc.nextLine();
```

```
    System.out.print("Enter New Marks: ");  
    double marks = sc.nextDouble();
```

```
    String query = "UPDATE student SET name = ?, course  
= ?, marks = ? WHERE roll_no = ?";
```

```
    PreparedStatement ps = con.prepareStatement(query);
```

```
    ps.setString(1, name);  
    ps.setString(2, course);  
    ps.setDouble(3, marks);  
    ps.setInt(4, rollNo);
```

```
    int rows = ps.executeUpdate();
```

```
    if (rows > 0) {  
        System.out.println("Student updated  
successfully");  
    } else {  
        System.out.println("Student not found");  
    }  
}
```

```
    ps.close();  
}
```

```
static void deleteStudent(Scanner sc) throws Exception {  
    System.out.print("Enter Roll Number to delete: ");  
    int rollNo = sc.nextInt();
```

```

        String query = "DELETE FROM student WHERE roll_no
= ?";

        PreparedStatement ps = con.prepareStatement(query);

        ps.setInt(1, rollNo);

        int rows = ps.executeUpdate();

        if (rows > 0) {
            System.out.println("Student deleted
successfully");
        } else {
            System.out.println("Student not found");
        }

        ps.close();
    }

    public static void main(String[] args) throws Exception {

        Class.forName("com.mysql.cj.jdbc.Driver");

        String db = "jdbc:mysql://localhost:3306/employee";
        String user = "root";
        String password = "admin@123";

        con = DriverManager.getConnection(db, user, password);

        Scanner sc = new Scanner(System.in);

        int choice;

        do {
            System.out.println("\n==== Student CRUD =====");
            System.out.println("1. Insert Student");
            System.out.println("2. Display Students");
            System.out.println("3. Update Student");
            System.out.println("4. Delete Student");
            System.out.println("5. Exit");

            System.out.print("Enter your choice: ");
            choice = sc.nextInt();

            try {
                switch (choice) {
                    case 1:
                        insertStudent(sc);
                        break;

```

```

        case 2:
            displayStudents();
            break;

        case 3:
            updateStudent(sc);
            break;

        case 4:
            deleteStudent(sc);
            break;

        case 5:
            System.out.println("Exiting...");
            break;

        default:
            System.out.println("Invalid choice");
    }
} catch (Exception e) {
    System.out.println("Error: " +
e.getMessage());
}

    } while (choice != 5);

    con.close();
    sc.close();
}
}

```

```

piyushrohokale@Piyushs-MacBook-Air JDBC % java -cp "src:lib/mysql-connector-j-26.7.0.jar" StudentCRUD
Enter Roll Number: 1
Enter Student Name: piyush
Enter Course: maths
Enter Marks: 99
Student inserted successfully

===== Student CRUD =====
1. Insert Student
2. Display Students
3. Update Student
4. Delete Student
5. Exit
Enter your choice: 3
Enter Roll Number to update: 1
Enter New Name: vyankatesh
Enter New Course: maths
Enter New Marks: 99
Student updated successfully

===== Student CRUD =====
1. Insert Student
2. Display Students
3. Update Student
4. Delete Student
5. Exit
Enter your choice: 2

Student Records
-----
Roll Number: 1
Name: vyankatesh
Course: maths
Marks: 99.0
-----

```

Github:

Java_assignments / assignment20 / 			Add file ▾		...
 VyankateshRohokale assignment 20			fd58d59 · now		 History
Name	Last commit message		Last commit date		
 ..					
 EmployeeCRUD.java	assignment 20		now		
 StudentCRUD.java	assignment 20		now		

Conclusion

The program successfully demonstrates CRUD operations on student records using Java JDBC and MySQL. It allows the user to insert, display, update, and delete student details such as roll number, name, course, and marks using PreparedStatement and SQL queries.